Perfect Maps

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Abstract—**Given positive integers r, s, u, and v, an $(r, s; u, v)$ Perfect Map (PM) is defined to be a periodic $r \times s$ binary array in which every $u \times v$ binary array appears exactly once as a periodic subarray. Perfect Maps are the natural extension of the de Bruijn sequences to two dimensions. In this paper the existence question for Perfect Maps is settled by giving constructions for Perfect Maps for all parameter sets subject to certain simple necessary conditions. Extensive use is made of previously known constructions by finding new conditions which guarantee their repeated application. These conditions are expressed as bounds on the linear complexities of the periodic sequences formed from the rows and columns of Perfect Maps.**

Index Terms—**Perfect Maps, de Bruijn array, window property, linear complexity.**

I. Introduction

GIVEN positive integers r, s, u, and v, an $(r, s; u, v)$ Perfect Map (PM) is defined to be a periodic $r \times s$ binary array in which every $u \times v$ binary array appears exactly once as a periodic subarray.

These arrays have applications in diverse areas of communications and coding, [1], such as two-dimensional range finding, scrambling, and various kinds of mask configurations. Perhaps the most obvious use for such arrays is in two-dimensional range finding (or position location). The basic idea is that, if an $(r, s; u, v)$ PM is written in some way onto a planar surface of dimensions r by s, then any device capable of examining a u by v rectangular subarray will be able to determine precisely its position on the surface. Brief mention is made of such an application in [2] and a more detailed description of applications of this type can be found in [3]. Note also that, in the one-dimensional case, similar position-detection applications have been suggested for de Bruijn and m-sequences by a number of authors (see for example, Bondy and Murty [4], Petriu and co-workers [5], [6], and Arazi [7]). To be useful in this type of application, the decoding problem, i.e., resolving the position of a particular subarray within a specified Perfect Map, is of great significance. This topic is addressed in [8].

Perfect Maps are also mathematically interesting in their own right as the natural extension of the binary de Bruijn sequences [9] to two dimensions. The fundamental question addressed in this paper is, given u and v, for which r and s does there exist an $(r, s; u, v)$ PM? In [8], Mitchell and this author gave three simple necessary conditions for the existence of an $(r, s; u, v)$ PM:

Result 1.1 ([18], Lemma 2.1): Suppose there exists an $(r, s; u, v)$ PM, A. Then

i) $rs = 2^{uv}$,
ii) $r > u$ or $r = u = 1$,
iii) $s > v$ or $s = v = 1$.

In view of Result 1.1 we will often write $r = 2^k$ and $s = 2^{uv-k}$, where $0 \le k \le uv$ and $2^k > u$, $2^{uv-k} > v$, so that we will refer to $(2^k, 2^{uv-k}; u, v)$ PMs.

The gap between these necessary conditions and the previously known constructions is wide. Reed and Stewart [2] gave an example of a $(4, 4; 2, 2)$ PM, and showed that in this simplest of cases there are essentially only two different PMs. Ma [10] and Fan *et al.* [11], using the terminology "de Bruijn arrays," noted that a PM with either r or s equal to 1 is simply a de Bruijn sequence of span u or v, respectively. They showed that given u and v, there exist r and s for which an $(r, s; u, v)$ PM can be constructed. They also gave two recursive constructions under which $(r, s; u, v)$ PMs satisfying certain constraints may be used to obtain $(r, 2^v s; u + 1, v)$ PMs and $(2r, 2^{v-1} s; u + 1, v)$ PMs. They justified repeated use of these constructions using ingenious special arguments and showed that there exists a $(2^{u^2/2}, 2^{u^2/2}; u, u)$ PM whenever u is even. In [1] Etzion gave a construction for a large class of Perfect Maps, settling the existence question for $(2^k, w^{uv-k}; u, v)$ PMs whenever $u < 2^k \le 2^u$ except when $u = k$ and $v = 2$. His constructions depend on the existence of Hamiltonian cycles in certain graphs; Etzion gave an algorithm for finding such cycles, while [8] gave an explicit construction for such cycles in terms of 2^k-ary de Bruijn and related sequences.

In this paper, we show that the conditions of Result 1.1 are in fact also sufficient for the existence of Perfect Maps. In other words, we settle the existence question for Perfect Maps. We make extensive use of the constructions of Fan *et al.*, but we give them more power by examining conditions which guarantee their repeated applicability. These conditions are expressed as upper bounds on the linear complexities of the rows and columns of Perfect Maps. We outline some basic notation and give the necessary background on linear complexity in Section II, while in Section III we give an analysis of the recursive constructions in [11], and an indication of how they might be used to construct large classes of Perfect Maps starting from a basic class. In Section IV, we review the work of Etzion in [1]; in particular, we study the linear complexi-

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ties of the sequences of the Perfect Factors which form the building blocks of Etzion's construction.

In Section V we give explicit examples of some special PMs with low u and v, which are of great use in settling the existence question for PMs with $2 \le u,\ v \le 5$ by means of a case by case discussion. The next step in our argument is to construct a suitable basic class of PMs to which the construction of [11] may be repeatedly applied; these constructions comprise Section VI. In the penultimate section, we complete our programme by using the basic class of PMs to produce PMs for all parameter sets satisfying the necessary conditions. Finally we discuss some related open problems in this area.

II. Basic Notation and Results

We follow the notation of [12], the results of which are fundamental to this paper. Let $S(p)$ denote the set of binary sequences satisfying $s_i = s_{i+p}$ for every $i \ge 0$. We say that the sequences of $S(p)$ have period p. If p is the least positive integer for which sequences (s) satisfies $s_i = s_{i+p}$, then we say that (s) has *least period* p. We may represent a member (s) of $S(p)$ by its *generating cycle*:

$$(s) = (s_0, s_1, \cdots, s_{p-1}).$$

Suppose that for all $i \ge 0$, (s) satisfies the linear recurrence

$$s_{i+m} + \sum_{j=1}^{m} a_j s_{i+m-j} = 0,$$

of degree m over the field $GF(2)$. Then if E denotes the *shift operator* whose action on (s) is defined by

$$Es_i = s_{i+1},$$

we have, for all $i \ge 0$,

$$\left(E^m + \sum_{j=1}^{m} a_j E^{m-j}\right) s_i = 0,$$

i.e.,

$$f(E)s_i = 0,$$

for a polynomial f of degree m. $f(E)$ is called the *minimal polynomial* of (s) when $f(E)s_i = 0$ and f has the least possible degree. In this case, the *linear complexity* of sequence (s) is defined to be the degree of the minimal polynomial of (s), and is denoted by $c(s)$. It can be shown that the minimal polynomial divides any other polynomial satisfying $g(E)s_i = 0$. Clearly when $(s) \in S(p)$, $(E^p + 1)s_i = 0$ and so $f(E) | E^p + 1$.

Suppose that $p = 2^k$. Then $E^{2^k} + 1 = (E + 1)^{2^k}$ because the field characteristic is 2, and so $f(E) = (E + 1)^{c(s)}$. This implies:

Result 2.1 ([12]): With notation as above, $c(s) = c$ if and only if $(E + 1)^{c-1} s_i = 1$, for all $i \ge 0$.

We define the *weight* of a sequence $(s) \in S(p)$ to be the sum $s_0 + s_1 + \cdots + s_{p-1}$. Note that the period p here need not be the least possible period of (s).

Result 2.2 ([12]): Suppose (s) has least period 2^k. Then

$$2^{k-1} + 1 \le c(s) \le 2^k$$

and $c(s) = 2^k$ if and only if (s) has odd weight.

Define the operator D by $D = E + 1$. Then for $(s) \in S(p)$,

$$D(s) = (s_0 + s_1, s_1 + s_2, \cdots, s_{p-2} + s_{p-1}, s_{p-1} + s_0).$$

In fact D is simply Lempel's homomorphism acting on periodic sequences—see [13] for details. We use the following result of Lempel:

Result 2.3 ([13]): Suppose $(s) \in S(p)$ has even weight. Then the set of preimages of (s) under D^{-1}, denoted by $D^{-1}(s)$, consists of a pair of complementary sequences of period p given by

$$D^{-1}(s) = \left\{ \left(0, s_0, s_0 + s_1, \cdots, \sum_{i=0}^{p-2} s_i\right), \left(1, 1 + s_0, 1 + s_0 + s_1, \cdots, 1 + \sum_{i=0}^{p-2} s_i\right) \right\}.$$

When (s) has odd weight, then $D^{-1}(s)$ contains one sequence of period $2p$ given by

$$D^{-1}(s) = \left\{ \left(0, s_0, s_0 + s_1, \cdots, \sum_{i=0}^{p-2} s_i, 1, 1 + s_0, 1 + s_0 + s_1, \cdots, 1 + \sum_{i=0}^{p-2} s_i\right) \right\}.$$

It is clear from the definition of D and Result 2.1 above that when p is a power of 2, an application of D to a sequence $(s) \in S(p)$ produces a sequence having linear complexity one less than that of (s), and conversely the sequence(s) in $D^{-1}(s)$ have linear complexity one greater than that of (s).

III. The Constructions of Fan, Fan, Ma, and Siu

In this section we examine the recursive constructions for Perfect Maps given in [11] and show how linear complexity of the sequences formed from rows and columns of Perfect Maps can be used as a tool to control the repeated application of these constructions.

Construction 3.1 ([11], Construction 5.1): Let A be an $(r, s; u, v)$ PM whose column weights are even (i.e., regarding the columns of A as s sequences of period r, each sequence has even weight). Then there exists a $(r, 2^v s; u + 1, v)$ PM B.

The construction proceeds by considering the columns of A as s binary sequences of period r and even weight, applying the operator D^{-1} to them to obtain a set of $2s$ sequences of period r and then using this new set as the columns of B in such a way as to ensure that every $(u + 1) \times v$ matrix appears once in B.

The ordering of columns is specified as follows. Let $(d) = (d_0, \cdots, d_{2^v - 1})$ be a span v binary de Bruijn sequence commencing with v zeros so that $(d)' = (d_1, \cdots, d_{2^v - 1})$ is the *punctured de Bruijn sequence* derived

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from (d) by deleting one of the initial run of zeros. Define a sequence (b) to have generating cycle

$$(b) = (0,\cdots,0,d_1,\cdots,d_{2^v-1},\cdots,d_1,\cdots,d_{2^v-1})$$

consisting of a run of s zeros followed by s copies of the sequence $(d)^\gamma$, so that (b) has least period $s + (2^v - 1)s = 2^v s$.

Then the array B is obtained by taking as its lth column the preimage under D^{-1} of the lth column of the array $[A|A|\cdots|A]$ (the array consisting of 2^v copies of A placed side by side), beginning with bit b_l. Thus, the first row of B is the sequence (b) and subsequent rows are obtained as the cumulative sum of (b) with the rows of $[A|A|\cdots|A]$.

A proof that Construction 3.1 yields a PM B with the desired parameters is to be found in [11].

Note that by Result 2.2, the column weights of A are even if and only if the linear complexities of the columns of A are nonmaximal, i.e., strictly less than the upper bound of Result 2.2. Also, applying D^{-1} to the columns of A results in each column of B having complexity one greater than the corresponding column of A. Thus the possibility of making, say, t repeated applications of Construction 3.1 depends on the original PM having all column complexities at least t less than the period. Clearly the determination of linear complexities is an important step in the construction of Perfect Maps. In what follows, we obtain bounds on the complexities of the rows and columns of the array B resulting from Construction 3.1. We begin with a lemma whose proof follows from the above discussion.

Lemma 3.2: Let A be a $(2^k, 2^{uv-k}; u, v)$ PM whose columns all have linear complexities less than or equal to c. Then if B is the PM obtained from A by using Construction 3.1, the columns of B all have linear complexities less than or equal to $c + 1$.

Lemma 3.3: Let A be a $(2^k, 2^{uv-k}; u, v)$ PM with even weight columns. Then if B is the $(2^k, 2^{(u+1)v-k}; u+1, v)$ PM obtained from A by using Construction 3.1, the rows of B all have equal linear complexities c satisfying

$$c \le 2^{(u+1)v-k} - 2^{uv-k}$$

(i.e., the linear complexities of the rows of B are at least 2^{uv-k} below the maximum possible complexity), unless $v = 1$ when the rows of B all have linear complexities equal to $2^{u-k} + 1$.

Proof: As observed above, the first row of B is the sequence (b) in Construction 3.1 which in turn is derived from the sequence $(d)^\gamma$. Consider for any $i \ge 0$, the expression

$$b_i + b_{i+s} + b_{i+2s} + \cdots + b_{i+(2^v-1)s}, \qquad (3.1)$$

where $s = 2^{uv-k}$.

Since $\gcd(s, 2^v - 1) = 1$ and there are s copies of $(d)^\gamma$ in a period of (b) we see that in the sum of (3.1) we pick up every term of $(d)^\gamma$ exactly once each, together with one zero from the run of s zeros. But the sequence $(d)^\gamma$ has even weight except in the case $v = 1$ where $(d)^\gamma = (1)$, and so we have

$$b_i + b_{i+s} + b_{i+2s} + \cdots + b_{i+(2^v-1)s} = 0 \qquad (v \ge 2).$$

Hence when $v \ge 2$, (b) satisfies a linear recurrence of degree $(2^v - 1)s$ and so

$$\begin{aligned} c(b) &\le (2^v - 1)s = 2^v 2^{uv-k} - 2^{uv-k} \\ &= 2^{(u+1)v-k} - 2^{uv-k}. \end{aligned}$$

When $v = 1$, $(b) = (0,\cdots,0,1,\cdots,1)$, a sequence consisting of two runs each of length 2^{u-k-1} and having linear complexity $2^{u-k} + 1$.

We now examine the other rows of B. They are obtained as the cumulative sum of (b) with the rows of $[A|A|\cdots|A]$; i.e., if B has jth row $(b)^j = (b_0^j, b_1^j, \cdots, b_{2^v s-1}^j)$ and $[A|A|\cdots|A]$ has jth row $(a)^j = (a_0^j, a_1^j, \cdots, a_{2^v s-1}^j)$, then

$$b_i^j = b_i + a_i^1 + \cdots + a_i^{j-1} \qquad (j \ge 1).$$

For $j \ge 1$ and any i,

$$\begin{aligned} (E+1)^{c(b)-1} b_i^j &= (E+1)^{c(b)-1}(b_i + a_i^1 + \cdots + a_i^{j-1}) \\ &= (E+1)^{c(b)-1} b_i + (E+1)^{c(b)-1} \\ &\quad \cdot (a_i^1 + \cdots + a_i^{j-1}). \end{aligned}$$

Now (b) has linear complexity $c(b)$ and least period $2^v 2^{uv-k}$, so by Result 2.2,

$$c(b) - 1 \ge 2^{v-1} 2^{uv-k} \ge 2^{uv-k} \qquad (\text{since } v \ge 1).$$

But the sequences $(a)^k$ have least period 2^{uv-k} or less and so $(E+1)^{c(b)-1} a_i^k = 0$ for all $i \ge 0$ and $k \ge 1$. Then

$$\begin{aligned} (E+1)^{c(b)-1} b_i^j &= (E+1)^{c(b)-1} b_i + (0 + \cdots + 0) \\ &= (E+1)^{c(b)-1} b_i \\ &= 1, \end{aligned}$$

and so by Result 2.1, $c((b)^j) = c(b)$ for all $j \ge 1$. The lemma follows. □

We note that if a PM A has even weight rows then Construction 3.1 may be applied to the rows of A to obtain a new PM whose row complexities are increased by 1 over those of A and whose column complexities are bounded by an expression similar to that in Lemma 3.3. In addition, if A is a $(2^k, 2^{uv-k}; u, v)$ PM with even column weights and B is the $(2^k, 2^{(u+1)v-k}; u+1, v)$ PM obtained from A by using Construction 3.1, then from Lemma 3.3, the rows of B all have equal linear complexities c satisfying

$$c \le 2^{(u+1)v-k} - 2^{uv-k}.$$

By Lemma 3.2 and Result 2.2, we may make between one and 2^{uv-k} applications of Construction 3.1 to the rows of B to obtain new Perfect Maps, each having column complexities that are nonmaximal and bounded by expressions similar to those of Lemma 3.3.

We next discuss a similar result to Construction 3.1 appearing in [11]:

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Construction 3.4 ([11], Construction 5.5): Let A be a $(r, s; u, v)$ PM having odd column sums (i.e., columns having maximal linear complexity). Then there exists a $(2r, 2^{v-1}s; u+1, v)$ PM, B.

Similarly to Construction 3.1, Construction 3.4 proceeds by considering the columns of A as s binary sequences of period r and maximal complexity, applying the operator D^{-1} to them to obtain a set of s sequences of period $2r$ (by Result 2.2) and then using this new set as the columns of B in such a way as to ensure that every $(u+1) \times v$ matrix appears once in B. The ordering of columns is specified as follows. Let (b) be a sequence of period $2^{v-1}s$ such that for each i $(1 \le i \le s)$, the 2^{v-1} v-tuples $b_{i+ts}, b_{i+1+ts}, \cdots, b_{i+v-1+ts}$ $(t = 0, 1, \cdots, 2^{v-1}-1)$ are all distinct and no two are complements of each other. Suitable sequences (b) are specified in [11] and in the proof below.

Then the array B is obtained by taking as its lth column either the preimage under D^{-1} of the lth column of the array $[A|A|\cdots|A]$ (2^{v-1} copies), or its complement, depending on whether bit b_l is 0 or 1. Thus the first row of B is the sequence (b) and subsequent rows are the corresponding cumulative sums.

We study the linear complexities of the rows and columns of a PM B resulting from Construction 3.4. Our result, analogous to Lemma 3.2 and Lemma 3.3, is

Lemma 3.5: Let A be a $(2^k, 2^{uv-k}; u, v)$ PM having odd column sums. Then there exist suitable sequences (b) such that when Construction 3.4 is used to produce a $(2^{k+1}, 2^{v-1}2^{uv-k}; u+1, v)$ PM B from A in conjunction with sequence (b), the linear complexities of the columns of B are equal to $2^k + 1$, the minimum possible, and if:

i) $v = 1$, then the linear complexity of each row of B is at most the greatest of the linear complexities of the rows of A.
ii) $v = 2$, then the row complexities of B are $2^{uv-k} + 2$.
iii) $v = 3$ or $v = 4$, then the row complexities of B are less than or equal to $2^{v-1}2^{uv-k} - 2^{uv-k} + 1$.
iv) $v \ge 5$, then the row complexities of B are less than or equal to $2^{v-1}2^{uv-k} - 2^{uv-k}$.

Proof: The linear complexities of the columns of B are one greater than those of A, as in Lemma 3.2, and A has column complexities equal to 2^k. It follows that B has column complexities equal to $2^k + 1$.

We now examine the complexities of the rows of B. When $v = 1$, a suitable sequence is $(b) = (0)$. Then the first row of B is (0) and the other rows of B are formed from the cumulative sum of the rows of A. It is easy to see that the linear complexity of each row of B is less than or equal to the greatest of the linear complexities of the rows of A, and (i) follows.

When $v = 2$, a suitable sequence is $(b) = (0, 0, \cdots, 0, 0, 1, 0, 1, \cdots, 0, 1)$ consisting of $s = 2^{uv-k}$ zeros followed by $s/2$ repetitions of the sequence (01). It is easy to see that (b) has linear complexity $s + 2$. A similar argument to that of Lemma 3.3 shows that the rows of B all have linear complexity equal to that of (b), and ii) follows.

For $v \ge 3$, we suggest the following procedure to define a suitable sequence (b) which is equivalent to that of [11]. Let (d) be a spand $v-1$ binary de Bruijn sequence beginning with $v-1$ zeros, and let (e) be the preimage under D^{-1} of (d) which begins with the all-zero v-tuple. Let $(e)'$ be the sequence obtained from (e) by deleting one of the zeros from the v-tuple and define sequence (b) to be the concatenation of s zeros and s copies of $(e)'$.

When $v \ge 5$, the result of [14] guarantees the existence of a spand $v-1$ binary de Bruijn sequence (d) having complexity $2^{v-2} + v - 1 \le 2^{v-1} - 2$. Then $c(e) = c(d) + 1 \le 2^{v-1} - 1$ so that (e) and hence $(e)'$ have even weight. An argument identical to that of Lemma 3.3 shows that $c(b) \le 2^{v-1}2^{uv-k} - 2^{uv-k}$. When $v = 3$ or $v = 4$, the procedure above results in sequences $(e)'$ with odd weight. Similar reasoning to that of Lemma 3.3 shows that in these cases, for every $i \ge 0$,

$$b_i + b_{i+s} + b_{i+2s} + \cdots + b_{i+(2^{v-1}-1)s} = 1.$$

Then

$$b_i + b_{i+1} + b_{i+s} + b_{i+s+1} + b_{i+2s} + b_{i+2s+1} + \cdots + b_{i+(2^{v-1}-1)s} + b_{i+(2^{v-1}-1)s+1} = 1 + 1 = 0,$$

so that $c(b) \le 2^{v-1}2^{uv-k} - 2^{uv-k} + 1$. A similar argument to that of Lemma 3.3 now shows that for each v the rows of B all have linear complexity equal to that of the appropriate (b). iii) and iv) follow. □

IV. Etzion's Construction

In [1], Etzion gave a construction for class of Perfect Maps with certain parameters. He introduces the idea of a Perfect Factor, a special collection of sequences forming the basis of his construction for Perfect Maps.

Result 4.1 ([1], Theorem 4): Let u, k be integers with $k \le u < 2^k$. Then there exists a collection of 2^{u-k} binary sequences, each of period 2^k, with the property that every binary u-tuple occurs in a unique sequence of the collection.

We denote such a collection as a (u, k) Perfect Factor. In view of the results of Section III, we study the linear complexities of the sequences of Etzion's Perfect Factors. Because of our need for sequences with nonmaximal linear complexities, we in fact adapt Etzion's construction to obtain the following:

Lemma 4.2: Let u, k be integers with $k \le u < 2^k$. Then there exists a (u, k) Perfect Factor whose sequences have linear complexities

i) $2^{k-1} + u$ if $k \le u < 2^{k-1}$,
ii) $u + 1$ if $2^{k-1} \le u < 2^k$.

Proof: The Perfect Factors of case i) are derived in the same way as those of [1], by repeatedly applying D^{-1} to a de Bruijn sequence of minimal complexity. For case ii) we also use the method of [1], where it is shown that the set of binary sequences of period 2^k and linear complexity $u + 1$ has the property that every binary u-tuple

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occurs exactly once as a subsequence of one sequence of the set. That is, this set constitutes a $(u, 2^k)$ Perfect Factor. We note that this class actually exists for the range $2^{k-1} \le u < 2^k$, whereas in the original proof of Result 4.1, the case where $u = 2^{k-1}$ was covered under i). □

From i) and ii) above it is clear that the sequences in a Perfect Factor can be chosen to have even weight except when $u = 2^k - 1$. It is easily shown that a Perfect Factor with even weight sequences cannot exist in this exceptional case.

We now give a brief review of the construction of [1], its relation to Perfect Factors, and Mitchell and Paterson's [8] version of Etzion's construction.

Result 4.3 ([1]): Suppose $u, v \ge 2$. Then there exist $(2^k, 2^{uv-k}; u, v)$ PM's for every k satisfying

$$u + 1 \le 2^k \le 2^u,$$

except when $v = 2$, where we require

$$u + 1 \le 2^k \le 2^{u-1}.$$

The basic idea behind this construction is to consider the graph whose vertices are the $2^k \times v$ binary matrices of the form $((s)^1, E^{t_1}(s)^2, \cdots, E^{t_{v-1}}(s)^v)$, where $(s)^i$ is a sequence in a (u, k) Perfect Factor, written vertically, and t_i is an integer between 0 and $2^k - 1$. Since there are 2^{u-k} distinct sequences in a (u, k) Perfect Factor, there are 2^{uv-k} distinct vertices in the graph. Two such vertices C and D are joined in the graph if the columns of D can all be shifted by the same amount in such a way that its first $v - 1$ columns agree with the last $v - 1$ columns of C. Etzion showed that these matrices can be linked up to form the desired PM by showing that a Hamiltonian cycle exists in the graph above.

The contribution of [8] was to give an explicit description of how to choose the relative shifts and the sequences of the Perfect Factor. This method has the advantage that an array so produced can be decoded more easily than the original Etzion array. That is, given a particular $u \times v$ matrix, it is possible to find the position of that matrix in the $r \times s$ PM. The construction is a generalization of Theorem 2 of [10], which deals with the case $k = u$ only. Lack of space prevents detailed discussion of the method of [8]; however, we will use it only in the simple case when $k = u$ in Section V below.

We now give a construction that produces PMs for most of Etzion's parameter sets, but which gives PMs having rows with linear complexities bounded below the maximum possible. This construction will be the first step in our construction of a basic class of PMs; the other steps will be completed in Section VI.

Theorem 4.4: Suppose u, v are integers with $u \ge 3$, $v \ge 2$, and suppose further that $u + 1 \le 2^k \le 2^{u-1}$ and if $k = u - 1$ then $v \ge 3$. Then there exists a $(2^k, 2^{uv-k}; u, v)$ PM whose row complexities are less than or equal to $2^{uv-k} - 2^{(u-1)v-k}$ and whose column complexities are

$$\begin{cases} 2^{k-1} + u & \text{if } k + 1 \le u < 2^{k-1} + 1, \\ u + 1 & \text{if } 2^{k-1} + 1 \le u < 2^k, \end{cases}$$

(so that the column weights are even unless $u = 2^{k-1}$ or $u = 2^k - 1$).

Proof: Because of the bounds on u, v, and k imposed in the statement of the theorem, Result 4.3 guarantees the existence of a $(2^k, 2^{(u-1)v-k}; u - 1, v)$ PM A. Since the sequences of a $(u - 1, k)$ Perfect Factor form the columns of the PM A it is clear from Lemma 4.2 that the columns of A can be chosen to have even weight. Hence we may apply Construction 3.1 to the columns of A to obtain a $(2^k, 2^{uv-k}; u, v)$ PM B having, in view of Lemma 3.2 and Lemma 3.3, row and column linear complexities as given in the statement of the theorem. □

V. Perfect Maps for Small u and v

Here we will establish the existence of $(2^k, 2^{uv-k}; u, v)$ PMs when $u, v \le 5$ for k subject to the conditions of Result 1.1 by means of a case by case discussion. This discussion is necessary primarily because it is difficult to extend the constructions of Section VI for our basic PMs below $u = 6$. Because of the repetitive nature of our arguments, we will not present every case here; the interested reader is referred to the Appendix for full details.

Throughout this section we make use of the fact that by transposing our arrays we may assume, without loss of generality, that $u \ge v$. It is sufficient then to construct arrays for every $u \ge v$ and k satisfying the conditions of Result 1.1. Moreover, when $u = v$ we need only construct PMs for $k \le u^2/2$. In fact, in this section we shall construct PMs for $2 \le v \le u \le 5$; PMs for $v = 1$ will be constructed in Section VI.

We begin by presenting four examples of Perfect Maps with low u and v whose row complexities and column complexities have special values. The properties of these special PM's will be used to construct certain other PMs in this section and in Section VII.

Lemma 5.1. There exists $(4, 16; 2, 3)$ PM A_1 with every column sum and row sum even. There exists $(4, 16; 2, 3)$ PM A_2 with every column sum even and every row sum odd.

Proof: Let (s) be the spand 2 binary de Bruijn sequence (0011), let $(t)^1$ be the 4-ary spand 2 de Bruijn sequence

$$(0022013211231033),$$

and let $(t)^2$ be the 4-ary spand 2 de Bruijn sequence

$$(3211220023013310).$$

Using the sequence (s) in conjunction with the sequences $(t)^1$ and $(t)^2$ in Theorem 2 of [10] or Construction 3.2 of [8], we obtain the following $(4, 16; 2, 3)$ PMs:

$$A_1 = \begin{pmatrix} 0001001010011111 \\ 0001000011010110 \\ 1110110101100000 \\ 1110111100101001 \end{pmatrix}$$

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and

$$A_2 = \begin{pmatrix} 0011010001001000 \\ 0101101110000010 \\ 1100101110110111 \\ 1010010001111101 \end{pmatrix}.$$

It may be verified that both A_1 and A_2 have even column sums (of linear complexity 3) and that the linear complexities of each of the rows of A_1 is 14. Also the rows of A_2 all have odd weight (and so their linear complexities are maximal). □

Lemma 5.2. There exists $(4, 64; 2, 4)$ PM A_3 with every column sum and row sum even. There exists $(4, 64; 2, 4)$ PM A_4 with every column sum even and every row sum odd.

Proof: Let (s) be the spand 2 binary de Bruijn sequence (0011), let $(t)^3$ be the 4 ary spand 3 de Bruijn sequence

(000202220012133201031322101312230113023211120323 1102123310030333)

and let $(t)^4$ be the 4-ary spand 3 de Bruijn sequence

(220103132223332320012133000303310021231102032111 2023011302210131).

Using the sequence (s) in conjunction with the sequences $(t)^3$ and $(t)^4$ in Theorem 2 of [10] or Construction 3.2 of [8], we obtain the following $(4, 64; 2, 4)$ PM's:

$$A_3 = \begin{pmatrix} 0000110100010001001101010111110111101100110011101000101000000011 \\ 0000110100001011000000010001010100111101011011011100110010001110 \\ 1111001011101110110010101000001000010011001100010111010111111100 \\ 1111001011110100111111101110101011000010100100100011001101110001 \end{pmatrix}$$

and

$$A_4 = \begin{pmatrix} 0100110101011001011101110000001111101010001110011001111011010000 \\ 0100000001010010011110100000111011100111001101001001001111011101 \\ 1011001010100110100010001111110000010101110001100110000100101111 \\ 1011111110101101100001011111000100011000110010110110110000100010 \end{pmatrix}$$

It may be verified that both A_3 and A_4 have even column sums (of linear complexity 3) and that the linear complexities of the rows of A_3 are all less than or equal to 57. Also the rows of A_4 all have odd weight (and so their linear complexities are maximal). □

Details of how the sequences $(t)^1$–$(t)^4$ were obtained are omitted. Linear complexities of rows were computed using the Games–Chan algorithm described in [15].

Next, we present constructions for PMs with $1 < v \leq u \leq 5$. We being with the case $u = 2$ where the only possible parameters are $(4, 4; 2, 2)$. It is shown in [2] that, up to cycling rows and columns, there are two possibilities:

$$\begin{pmatrix} 0010 \\ 0001 \\ 0111 \\ 1011 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0001 \\ 0010 \\ 1011 \\ 0111 \end{pmatrix},$$

both having odd row and column sums.

When $u = 3$ and $v = 2$, we have the following parameter sets:

i) $(2^2, 2^4; 3, 2)$: a $(4, 16; 3, 2)$ PM may be constructed using the method of [1]. A useful example is the following PM whose rows have linear complexities equal to 14:

$$\begin{pmatrix} 0011001000110111 \\ 0011000101100001 \\ 0011011100110010 \\ 1100101110011011 \end{pmatrix}.$$

ii) $(2^3, 2^3; 3, 2)$: an $(8, 8; 3, 2)$ PM appears in [11], Example 5.6. It has row complexities equal to 6 and column complexities equal to 5.

iii) $(2^4, 2^2; 3, 2)$: suitable PM's are the transpose of A_1 and A_2 of this section.

Similarly when $u = 3$ and $v = 3$, we have

i) $(2^2, 2^7; 3, 3)$: apply Construction 3.1 to the columns of the PM A_1 to obtain a PM with the desired parameters having row complexities less than or equal to $2^7 - 2^4$ by Lemma 3.3.

ii) $(2^3, 2^6; 3, 3)$: a suitable PM may be obtained by applying Construction 3.1 to the rows of the $(2^3, 2^3; 3, 2)$ PM above and then transposing. Lemma 3.3 guarantees that the row complexities will be less than or equal to $2^5 - 2^3$.

iii) $(2^4, 2^5, 3, 3)$: a suitable PM may be obtained by applying Construction 3.1 to the rows of the $(2^2, 2^4; 3, 2)$ PM above and then transposing. The resulting PM has column complexities 15 and row complexities less than or equal to $2^5 - 2^2$.

We claim that PMs for $u = 4$ and $u = 5$ can be constructed in a similar fashion to those above by combining the use of Construction 3.1, Construction 3.4, and Theorem 4.4 with the PMs A_1, A_2, A_3, A_4 and the examples already obtained, while keeping track of the linear complexities of rows and columns. The full details are given in the Appendix.

VI. Construction of a Basic Class of Perfect Maps

We are now ready to supplement the result of Theorem 4.4 with further constructions. Our basic class is to consist of PMs with parameters $(2^k, 2^{uv-k}; u, v)$, where $6 \leq u < 2^k < u2^u$ and $v \leq u$ with a bound on row complexities; we will repeatedly apply Construction 3.1 to the rows of

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PMs in this class in Section VII and the reason for desiring such a range of parameters will become clear there.

On examining Theorem 4.4, the reader will notice that we have already constructed PMs with the desired parameters for a large portion of the range above. Specifically, the parameter sets still missing are those with

i) $v = 1$,
ii) $k = u - 1$ and $v = 2$,
iii) $k = u$ and $v \geq 2$,
iv) $u < k < u + \log_2(u + 1)$ and $v \geq 2$.

For case i), PMs with $v = 1$ can easily be constructed using a (u, k) Perfect Factor; however, we are interested in PMs with low row complexities, and the simple idea of ordering the sequences of such a factor as the columns of a PM fails to give the required complexities. It is readily verified from Result 1.1 that for $u \leq 4$, every PM with $v = 1$ has at most two columns. It is also simple to show that a PM with $v = 1$ having at most two columns must have at least one odd weight row. Conversely, in the next construction, we construct PMs for $v = 1$ having minimal row complexities for every $k \leq u - 2$ whenever $u \geq 5$.

Lemma 6.1: Suppose $u \geq 5$ and $k \leq u - 2$. Then there exists a $(2^k, 2^{u-k}, u, 1)$ PM having rows with linear complexity equal to $2^{u-k-1} + 1$.

Proof: We aim to produce a $(2^k, 2^{u-k}; u, 1)$ PM. Then by Result 1.1 we require $2^k > u$, so $2^k \geq (u - 1) + 2$. Then from Lemma 4.2, there exists a $(u - 1, k)$ Perfect Factor whose sequences have period 2^k and linear complexity at most $2^k - 1$. Using these sequences in any order to form the columns of an array, we obtain a $(2^k, 2^{u-k-1}; u - 1, 1)$ PM A with even column sums. Applying Construction 3.1 to the columns of A, we obtain a $(2^k, 2^{u-k}; u, 1)$ PM B, with row complexities $2^{u-k-1} + 1$ according to Lemma 3.3. □

In the case $k =$ and $k = u - 1$ not covered by Lemma 6.1, PMs with the desired parameters may be obtained by using a de Bruijn sequence in the former case and by using the sequences of a $(u, u - 1)$ Perfect Factor as the column of an array in the latter.

We next address ii) above by giving a cnstruction for PMs with $k = u - 1$ and $v = 2$ having low row complexities.

Lemma 6.2: Suppose $u \geq 4$. Then there exists a $(2^{u-1}, 2^{u+1}; u, 2)$ PM having row complexities equal to $2^u + 2$.

Proof: Let (s) be a spand $u - 2$ de Bruijn sequence having linear complexity $2^{u-2} - 1$ (such a sequence always exists; for example in [12] it is shown that a de Bruijn sequence derived from an m-sequence has this linear complexity). Applying the operator D^{-1} to (s), we obtain a $(u - 1, u - 2)$ Perfect Factor consisting of two sequences of period and linear complexity 2^{u-2}. Thus the sequences of the Perfect Factor have odd weight.

Use this Perfect Factor in Result 4.3 to obtain a $(2^{u-2}, 2^u; u - 1, 2)$ PM A with odd column sums. Applying Construction 3.4 to the columns of A and using Lemma 3.5, we obtain a $(2^{u-1}, 2^{u+1}; u, 2)$ PM B having minimal column complexities and row complexities equal to $2^u + 2$. □

Our next theorem deals with the case $k = u$ in iii) above.

Theorem 6.3: Suppose $1 < v \leq u$ and $u \geq 6$. Then there exists a $(2^u; 2^{uv-u}; u, v)$ PM having row complexities less than or equal to $2^{uv-u} - 2^{(u-1)v-u}$.

Proof: The proof is split into cases for u even and odd. We begin with u even and construct PM's having $k = u$ for $u = 6$ and $1 < v \leq 6$.

The transpose of PM A_3 is a $(2^6, 2^2; 4, 2)$ PM having column complexities 57 or less. Apply Construction 3.1 to the columns of this PM twice to obtain a $(2^6, 2^6; 6, 2)$ PM A_5 having row complexities less than or equal to $2^6 - 2^4$.

From Section V, there exists a $(2^3, 2^6; 3, 3)$ PM having row complexities at least 2^3 less than the maximum. Apply Construction 3.1 three times to the rows of this PM and transpose to obtain a $(2^6, 2^{12}; 6, 3)$ PM with row complexities less than or equal to $2^{12} - 2^9$.

Again from Section V, there exists a $(2^3, 2^6; 3, 3)$ PM having row complexities at least 2^3 less than the maximum. Applying Construction 3.1 once to the rows and then three times to the columns, we obtain a $(2^6, 2^{18}; 6, 4)$ PM with row complexities less than or equal to $2^{18} - 2^{14}$.

Recall that A_1 is a $(2^2, 2^4; 2, 3)$ PM with row complexities equal to 14. Applying Construction 3.1 twice to the rows, we obtain a $(2^6, 2^4; 2, 5)$ PM with column complexities less than or equal to $2^6 - 2^4$. Using Construction 3.1 four times on the columns of this PM, we obtain a $(2^6, 2^{24}; 6, 5)$ PM with row complexities less than or equal to $2^{24} - 2^{19}$.

Applying Construction 3.1 four times to the rows of the $(2^6, 2^6; 6, 2)$ PM A_5 above, we obtain a $(2^{30}, 2^6; 6, 6)$ PM with column complexities less than or equal to $2^{30} - 2^{24}$. Transposing, we obtain a $(2^6, 2^{30}; 6, 6)$ PM with row complexities less than or equal to $2^{30} - 2^{24}$.

Still assuming u even, suppose $u \geq 8$ and $v \geq 2$. Write $u = 2s + 2$, so that $s \geq 3$. Then $v - 1 \geq 1$ and $2^s \geq s + 3$, so that by Theorem 4.4 when $v - 1 \geq 2$ and by Lemma 6.1 when $v - 1 = 1$, there exists a $(2^s, 2^{(s+2)(v-1)-s}; s + 2, v - 1)$ PM A having even row sums. Apply Construction 3.1 to the rows of A to obtain a $(2^{2^s+2}, 2^{(s+2)(v-1)-s}; s + 2, v)$ PM B having column complexities less than or equal to $2^{2^s+2} - 2^s$ by Lemma 3.3.

For $s > 1$, $2^s > s$ and so we may repeatedly apply Construction 3.1 to the columns of B a total of s times to obtain a $(2^u, 2^{uv-k}; u, v)$ PM C having column complexities less than or equal to $2^u - 2^s + s$ and row complexities less than or equal to $2^{uv-k} - 2^{(u-1)v-k}$.

This completes the cases when u is even. Next we consider u odd and $u \geq 7$. We initially deal with the case $v = 2$ and consider two subcases. First, suppose $u = 7$, and consider the $(2^6, 2^2; 4, 2)$ PM obtained by transposing A_4. It has odd weight columns and so on applying Construction 3.4 to its columns we obtain a $(2^7, 2^3; 5, 2)$ PM having column complexities $2^6 + 1$. Applying Construction 3.1 twice to the columns of this PM, we obtain a

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$(2^7, 2^7; 7, 2)$ PM having row complexities less than or equal to $2^7 - 2^5$.

Second, suppose $u \geq 9$. Write $u = 2s + 3$ so $s \geq 3$ and $2^s \geq s + 4$. Then by Lemma 6.1, there exists a $(2^s, 2^3; s + 3, 1)$ PM A with even row sums. Apply Construction 3.1 to the rows of A, then s times to the columns of the resulting $(2^{s+s+3}, 2^3; s + 3, 2)$ PM to obtain the desired $(2^u, 2^u; u, 2)$ PM having row complexities less than or equal to $2^u - 2^{u-2}$.

Last, we consider cases where u is odd and $v \geq 3$. Suppose $u \geq 7$ and $v \geq 3$. Write $u = 2s + 1$. Then $v - 1 \geq 2$, $s + 1 \geq 4$, and $2^s \geq s + 2$, so that by Theorem 4.4 when $v - 1 \geq 3$ and by Lemma 6.2 when $v - 1 = 2$, there exists a $(2^s, 2^{(s+1)(v-1)-s}; s + 1, v - 1)$ PM A having even row sums. Apply Construction 3.1 to the rows of A to obtain a $(2^{s+s+1}, 2^{(s+1)(v-1)-s}; s + 1, v)$ PM B having column complexities at most $2^{s+s+1} - 2^s$. Now $2^s > s$, so we may repeatedly apply Construction 3.1 to the columns of B a total of s times to obtain a $(2^u, 2^{uv-k}; u, v)$ PM C having column complexities less than or equal to $2^u - 2^s + s$ and row complexities less than or equal to $2^{uv-k} - 2^{(u-1)v-k}$. □

Our final construction in this section deals with the remaining parameter sets in iv) above.

Theorem 6.4: Suppose $u \geq 6$ and $v \geq 2$. Let $0 < w < \log_2(u + 1)$. Then there exists a $(2^{u+w}, 2^{uv-u-w}; u, v)$ PM with row complexities less than or equal to $2^{uv-u-w} - 2^{(u-1)v-u-w}$.

Proof: Again the proof is split into two cases depending on whether u is even or odd. We suppose first that u is odd, so $u \geq 7$ and $v \geq 2$. Write $u = 2s + 1$, so that $s \geq 3$, and suppose $0 < w < \log_2(u + 1)$. It is easy to verify that $2^s \geq s + w + 1$ and $2^{(s+w+1)(v-1)-s} > v - 1$. By Theorem 4.4 or Lemma 6.1 in the case $v = 2$, there exists a $(2^s, 2^{(s+w+1)(v-1)-s}; s + w + 1, v - 1)$ PM A having even row sums.

Applying Construction 3.1 to the rows of A, we obtain a $(2^{2s+w+1}, 2^{(s+w+1)(v-1)-s}; s + w + 1, v)$ PM B with column complexities less than or equal to $2^{2s+w+1} - 2^s$. For $s \geq 3$, $2^s > s$ and so we may apply Construction 3.1 $s - w$ times to the columns of B to obtain a $(2^{u+w}, 2^{uv-u-w}; u, v)$ PM having row complexities less than or equal to $2^{uv-u-w} - 2^{(u-1)v-u-w}$. Also, since we make strictly less than 2^s applications of Construction 3.1, we can deduce that this PM has even weight columns.

Suppose now that u is even, $u \geq 6$, and $v \geq 2$. Write $u = 2s$ and suppose $0 < w < \log_2(u + 1)$. Then $s \geq 3$, $2^s \geq s + w$, and $2^{(s+w)(v-1)-s} > v - 1$. By Theorem 4.4, or Lemma 6.1 in the case $v - 1 = 1$, $w > 1$, and Lemma 6.2 in the case $v - 1 = 2$, $w = 1$, there exists a $(2^s, 2^{(s+2)(v-1)-s}; s + w, v - 1)$ PM A with even row sums.

Applying Construction 3.1 to the rows of A, we obtain a $(2^{2s+2}, 2^{(s+w)(v-1)-s}; s + w, v)$ PM B with column complexities less than or equal to $2^{2s+w} - 2^s$. For $s \geq 3$, $2^s > s$ and so we may apply Construction 3.1 $s - w$ times to the columns of B to obtain a $(2^{u+w}, 2^{uv-u-w}; u, v)$ PM having row complexities less than or equal to $2^{uv-u-w} - 2^{(u-1)v-u-w}$.

In the remaining even case where $v - 1 = 1$, $w = 1$, we use a special argument. When $u = 6$, consider the $(2^6, 2^2; 4, 2)$ PM obtained by transposing A_4. It has odd weight columns and so on applying Construction 3.4 to its columns we obtain a $(2^7, 2^3; 5, 2)$ PM having column complexities $2^6 + 1$. Applying Construction 3.1 to the columns of this PM, we obtain a $(2^7, 2^5; 6, 2)$ PM having row complexities less than or equal to $2^5 - 2^3$. When $u \geq 8$, from the odd case above, there exists a $(2^{u+1}, 2^{u-3}; u - 1, 2)$ PM A having even weight columns. Applying Construction 3.1 to the columns of A, we obtain a $(2^{u+1}, 2^{u-1}; u, 2)$ PM with row complexities less than or equal to $2^{u-1} - 2^{u-3}$. This completes the proof of the theorem. □

We summarize the results of this section in the following:

Theorem 6.5: For every $u \geq 6$ there exists a $(2^k, 2^{u-k}; u, 1)$ PM with row complexities equal to $2^{u-k-1} + 1$ for every k with $u + 1 \leq 2^k \leq 2^{u-2}$. For every $u \geq 6$ and every $2 \leq v \leq u$, there exists a $(2^k, 2^{uv-k}; u, v)$ PM with row complexities less than or equal to $2^{uv-k} - 2^{(u-1)v-k}$ for every k with $u + 1 \leq 2^k < u2^u$, i.e., with $\lceil \log_2(u + 1) \rceil \leq k < u + \lceil \log_2(u + 1) \rceil$.

VII. Existence of Perfect Maps for Every Possible Parameter Set

In this section we use the basic class of PMs of Theorem 6.5 to construct PMs for every possible parameter set satisfying the necessary conditions of Result 1.1. In the following theorem we study the problem of general u and v with $u \geq 6$, $1 < v \leq u$. Then we combine these studies with Lemma 6.1 and the results of Section V to produce our last result.

Theorem 7.1. Suppose $u \geq 6$ and $1 < v \leq u$. Then there exist $(2^k, 2^{uv-k}; u, v)$ PMs for every set of parameters satisfying Result 1.1.

Proof: Suppose $u \geq 6$ and $1 < v \leq u$. From Result 1.1 we see that we must construct $(2^k, 2^{uv-k}; u, v)$ PMs for every k with $u < 2^k$ and $2^{uv-k} > v$. Writing $k = w + ut$ where $\lceil \log_2(u + 1) \rceil \leq w < u + \lceil \log_2(u + 1) \rceil$ and $0 \leq t \leq v - 1$, we split the proof into two cases: $0 \leq t \leq v - 2$ and $t = v - 1$.

Suppose $0 \leq t \leq v - 2$, so $2 \leq v - t \leq v$. Then from our choice for the range of w, $2^w > u$. Moreover

$$u(v - t) - w \geq 2u - u - \lceil \log_2(u + 1) \rceil + 1 = u + 1 - \lceil \log_2(u + 1) \rceil,$$

and for $u \geq 6$

$$2^{u+1-\lceil \log_2(u+1) \rceil} > u \geq v \geq v - t.$$

Then by Theorem 6.5 there exists a $(2^w, 2^{u(v-t)-w}; u, v - t)$ PM A with row complexities less than or equal to $2^{u(v-t)-w} - 2^{(u-1)(v-t)-w}$. Now

$$(u - 1)(v - t) - w \geq 2(u - 1) - u - \lceil \log_2(u + 1) \rceil + 1 = u - 1 - \lceil \log_2(u + 1) \rceil,$$

and for $u \geq 6$

$$2^{u-1-\lceil \log_2(u+1) \rceil} \geq u - 2 \geq v - 2 \geq t.$$

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Hence the row complexities of A are at least t below the maximum possible value and so we may make t applications of Construction 3.1 to the rows of A to obtain a $(2^{w+ut}, 2^{u(v-t)-w}; u, v)$ PM, i.e., a $(2^{w+ut}, 2^{uv-(w+ut)}; u, v)$ PM. (Note that when $t = 0$ there is nothing to be done; A is a suitable PM.)

In the second case we suppose $t = v - 1$ and we may assume that $\lceil \log_2 (u + 1) \rceil \leq w \leq u$. We aim to produce a PM having 2^{u-w} columns and so from Result 1.1, we require

$$2^{u-w} \geq v + 1. \tag{7.1}$$

When $w = u$ or $w = u - 1$, this inequality is not satisfied (since $v > 1$), and so no such PM can exist. We may therefore assume that $u - w \geq 2$ and so by Theorem 6.5 there exists a $(2^w, 2^{u-w}; u, 1)$ PM A with row complexities equal to $2^{u-w-1} + 1$. We wish to make $t = v - 1$ applications of Construction 3.1 to the rows of A, so we require

$$2^{u-w} - (2^{u-w-1} + 1) \geq v - 1,$$

i.e.,

$$2^{u-w-1} \geq v. \tag{7.2}$$

If $2^{u-w} \geq 2v$, then (7.2) holds and we proceed to make t applications of Construction 3.1 to the rows of A to obtain the required $(2^{uv-u+w}, 2^{u-w}; u, v)$ PM. Otherwise, from (7.1) we have

$$v + 1 \leq 2^{u-w} < 2v \leq 2^v \quad \text{for } v > 1,$$

and the desired $(2^{uv-u+w}, 2^{u-w}; u, v)$ PM may be obtained from the transpose of a PM from Result 4.3.

This completes the proof for $t = v - 1$, the second of our two cases. □

Theorem 7.2: The conditions of Result 1.1 on u, v, and $r = 2^k$, $s = 2^{uv-k}$ are sufficient for the existence of a $(2^k, 2^{uv-k}; u, v)$ Perfect Map.

Proof: Suppose without loss of generality that $u \geq v$. Construction of Perfect Maps for $v = 1$ and every possible $r = 2^k$ for each u were given in Lemma 6.1 so we may assume that $v > 1$. When $u \leq 5$ the results of Section V show the existence of PM's for each $v \leq u$ and every $r = 2^k$, $s = 2^{uv-k}$ satisfying the conditions of Result 1.1. When $u \geq 6$ and $v > 1$, Theorem 7.1 may be used to construct PM's for every $r = 2^k$, $s = 2^{uv-k}$ satisfying the conditions of Result 1.1. □

VIII. Related Open Problems

It has already been noted that Perfect Maps are the two-dimensional generalizations of binary de Bruijn sequences. The existence (and indeed number) of such sequences has long been known, even in the nonbinary case. We can ask for which parameter sets do Perfect Maps exist in the more general c-ary case? Necessary conditions similar to those of Result 1.1 can be obtained. We conjecture that these are also sufficient. Also, we have constructed many Perfect Maps for each parameter set (we do not attempt to enumerate how many here); it would be of interest to give an exact expression for the number of Perfect Maps for each parameter set, analogous to the result in [9] for binary de Bruijn sequences. This number is known in the simplest case: there are two PM's with parameters $(4, 4; 2, 2)$—details may be found in [2].

Higher-dimensional arrays analogous to Perfect Maps and de Bruijn sequences can be defined. There are some existence results for the three-dimensional generalization in [16]. A complete solution would be of doubtful practical significance, but mathematically interesting.

Appendix

In this appendix we give constructions for PMs with $1 < v \leq u$ for $u = 4$ and $u = 5$, these being the parameter sets omitted in Section V.

A. The case $u = 4$

1) The case $u = 4$, $v = 2$

i) $(2^3, 2^5; 4, 2)$: apply Construction 3.1 to the columns of the $(2^3, 2^3; 3, 2)$ PM of Section V to obtain a PM with the desired parameters having columns with linear complexities 6 and rows with linear complexities less than or equal to $2^5 - 2^3$.

ii) $(2^4, 2^4; 4, 2)$: apply Construction 3.1 to the rows of the PM A_1 of Section V and transpose to obtain a PM with the desired parameters having columns with linear complexities 15 and rows with linear complexities less than or equal to $2^4 - 2^2$.

iii) $(2^5, 2^3; 4, 2)$: apply Construction 3.4 to the rows of the PM A_2 of Section V and transpose to obtain a PM with the desired parameters having columns with linear complexities $2^4 + 1$ and rows with linear complexities $2^2 + 2$.

iv) $(2^6, 2^2; 4, 2)$: the PMs A_3 and A_4 of Section V may be transposed to obtain PMs with the desired parameters with columns with linear complexities either less than or equal to 57 or maximal and with rows with linear complexities equal to 3.

2) The case $u = 4$, $v = 3$

i) $(2^3, 2^9; 4, 3)$: apply Construction 3.1 to the columns of the $(2^3, 2^6; 3, 3)$ PM of Section V to obtain a PM with the desired parameters having columns with linear complexities 8 and rows with linear complexities less than or equal to $2^9 - 2^6$.

ii) $(2^4, 2^8; 4, 3)$: apply Construction 3.1 to the columns of the $(2^4, 2^5; 3, 3)$ PM of Section V to obtain a PM with the desired parameters having columns with linear complexities 16 and rows with linear complexities less than or equal to $2^8 - 2^5$.

iii) $(2^5, 2^7; 4, 3)$: apply Construction 3.1 to the rows of the $(2^4, 2^5; 3, 3)$ PM of Section V and transpose to obtain a PM with the desired parameters having columns with linear complexities less than or equal to $2^5 - 2^2 + 1$ and rows with linear complexities less than or equal to $2^7 - 2^4$.

iv) $(2^6, 2^6; 4, 3)$: apply Construction 3.1 to the rows of the $(2^3, 2^6; 3, 3)$ PM of Section V and transpose to obtain a PM with the desired parameters having columns with linear complexities less than or equal to $2^6 - 2^3 + 1$ and rows with linear complexities less than or equal to $2^6 - 2^3$.

v) $(2^7, 2^5; 4, 3)$: apply Construction 3.1 to the rows of the PM of A.1i). The resulting PM has columns with linear complexities less than or equal to $2^7 - 2^3$ and rows with linear complexities less than or equal to $2^5 - 2^3 + 1$.

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vi) $(2^8, 2^4; 4, 3)$: apply Construction 3.1 to the rows of the PM of A.1ii). The resulting PM has columns with linear complexities less than or equal to $2^8 - 2^4$ and rows with linear complexities less than or equal to 13.

vii) $(2^9, 2^3; 4, 3)$: apply Construction 3.1 to the rows of the PM of A.1iii). The resulting PM has columns with lienar complexities less than or equal to $2^9 - 2^5$ and rows with linear complexities equal to 7.

viii) $(2^{10}, 2^2; 4, 3)$: apply Construction 3.1 to the rows of the PM of A.1iv) having even row sums. The resulting PM has columns with linear complexities less than or equal to $2^{10} - 2^6$ and rows with linear complexities less than or equal to 58.

3) The case $u = 4$, $v = 4$

PMs for $u = 4$ and $v = 4$ with $3 \le k \le 8$ having rows with linear complexities less than or equal to $2^{16-k} - 2^{12-k}$ may be obtained by applying Construction 3.1 to the rows of the corresponding $(2^{12-k}, 2^k; 4, 3)$ PMs in A.2) above and transposing. The resulting PMs have columns with linear complexities one greater than the linear complexities of the rows of the corresponding PMs of A.2).

B. The case $u = 5$

1) The case $u = 5$, $v = 2$

i) $(2^3, 2^7; 5, 2)$: apply Construction 3.1 to the columns of the PM of A.1i). The resulting PM has rows with linear complexities less than or equal to $2^7 - 2^5$.

ii) $(2^4, 2^6; 5, 2)$: apply Construction 3.1 to the columns of the PM of A.1ii). The resulting PM has rows with linear complexities less than or equal to $2^6 - 2^4$.

iii) $(2^5, 2^5; 5, 2)$: apply Construction 3.1 to the columns of the PM of A.1iii). The resulting PM has rows with linear complexities less than or equal to $2^5 - 2^3$ and columns with linear complexities $2^4 + 2$.

iv) $(2^6, 2^4; 5, 2)$: apply Construction 3.1 to the rows of the PM A_3 of Section V and transpose to obtain a PM with the desired parameters and rows with linear complexities less than or equal to $2^4 - 2^2$.

v) $(2^7, 2^3; 5, 2)$: apply Construction 3.4 to the rows of the PM A_4 of Section V and transpose to obtain a PM with the desired parameters having columns with linear complexities equal to $2^6 + 1$ and rows with linear complexities equal to 6.

vi) $(2^8, 2^2; 5, 2)$: such a PM may be obtained by transposing a $(2^2, 2^8; 2, 5)$ PM from Result 4.3.

2) The case $u = 5$, $v = 3$

i) $(2^3, 2^{12}; 5, 3)$: a PM with these parameters and rows with linear complexities less than or equal to $2^{12} - 2^9$ may be derived from Theorem 4.4.

ii) $(2^4, 2^{11}; 5, 3)$: apply Construction 3.1 to the columns of the PM of A.2ii) to obtain a PM with the required parameters having rows with linear complexities less than or equal to $2^{11} - 2^8$.

iii) $(2^5, 2^{10}; 5, 3)$: apply Construction 3.1 to the columns of the PM of A.2iii) to obtain a PM with the required parameters having columns with linear complexities less than or equal to $2^5 - 2^2 + 2$ and rows with linear complexities less than or equal to $2^{10} - 2^7$.

iv) $(2^6, 2^9; 5, 3)$: apply Construction 3.1 to the columns of the PM of A.2iv) to obtain a PM with the required parameters having columns with linear complexities less than or equal to $2^6 - 2^3 + 2$ and rows with linear complexities less than or equal to $2^9 - 2^6$.

v) $(2^7, 2^8; 5, 3)$: apply Construction 3.1 to the columns of the PM of A.2v) to obtain a PM with the required parameters having columns with linear complexities less than or equal to $2^7 - 2^4 + 2$ and rows with linear complexities less than or equal to $2^8 - 2^5$.

vi) PM's for $u = 5$ and $v = 3$ with $8 \le k \le 12$ having rows with even weights may be obtained by applying Construction 3.1 to the rows of the corresponding $(2^{k-5}, 2^{15-k}; 5, 2)$ PMs in B.1) above.

vii) $(2^{13}, 2^2; 5, 3)$: such a PM may be obtained from Result 4.3.

3) The case $u = 5$, $v = 4$

i) $(2^3, 2^{17}; 5, 4)$: a PM with these parameters having rows with linear complexities less than or equal to $2^{17} - 2^{13}$ may be obtaned from Theorem 4.4.

ii) $(2^4, 2^{16}; 5, 4)$: a PM with these parameters having rows with linear complexities less than or equal to $2^{16} - 2^{12}$ may be obtained from Theorem 4.4.

iii) $(2^5, 2^{15}; 5, 4)$: apply Construction 3.1 to the columns of the $(2^5, 2^{11}; 4, 4)$ PM of A.3) to obtain a PM with the required parameters having columns with linear complexities less than or equal to $2^5 - 2^3 + 3$ and rows with linear complexities less than or equal to $2^{15} - 2^{11}$.

iv) $(2^6, 2^{14}; 5, 4)$: apply Construction 3.1 to the columns of the $(2^6, 2^{10}; 4, 4)$ PM of A.3) to obtain a PM with the required parameters having linear complexities less than or equal to $2^6 - 2^3 + 2$ and rows with linear complexities less than or equal to $2^{14} - 2^{10}$.

v) $(2^7, 2^{13}; 5, 4)$: apply Construction 3.1 to the columns of the $(2^7, 2^9; 4, 4)$ PM of A.3) to obtain a PM with the required parameters having columns with linear complexities less than or equal to $2^7 - 2^4 + 2$ and rows with linear complexities less than or equal to $2^{13} - 2^9$.

vi) PMs for $u = 5$ and $v = 4$ with $8 \le k \le 12$ having rows with even weight may be obtained by applying Construction 3.1 to the rows of the corresponding $(2^{k-5}, 2^{20-k}; 5, 3)$ PMs in B.2) above.

vii) Similarly, PMs for $u = 5$ and $v = 4$ with $13 \le k \le 17$ having rows with even weight (except when $k = 17$) may be obtained by applying Construction 3.1 twice to the rows of the corresponding $(2^{k-10}, 2^{20-k}; 5, 2)$ PM's in B.1) above.

4) The case $u = 5$, $v = 5$

i) $(2^3, 2^{22}; 5, 5)$: a PM with these parameters may be obtained from Result 4.3.

ii) $(2^4, 2^{21}; 5, 5)$: a PM with these parameters may be obtained from Result 4.3.

iii) $(2^5, 2^{20}; 5, 5)$: a PM with these parameters may be obtained from Result 4.3.

iv) $(2^6, 2^{19}; 5, 5)$: apply Construction 3.1 to the rows of the $(2^{14}, 2^6; 5, 4)$ PM of B.3) and transpose to obtain a PM with the required parameters.

v) $(2^7, 2^{18}; 5, 5)$: apply Construction 3.1 to the rows of the $(2^{13}, 2^7; 5, 4)$ PM of B.3) and transpose to obtain a PM with the required parameters.

vi) PMs for $u = v = 5$ with $8 \le k \le 12$ may be obtained by applying Construction 3.1 to the rows of the corresponding $(2^{k-5}, 2^{25-k}; 5, 4)$ PMs in B.3) above.

This completes the constructions required to prove the existence of PMs with $1 < v \le u$ for $u = 4$ and $u = 5$. Note that we

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have constructed PMs with rows and columns having linear complexities often considerably below the maximal values.

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